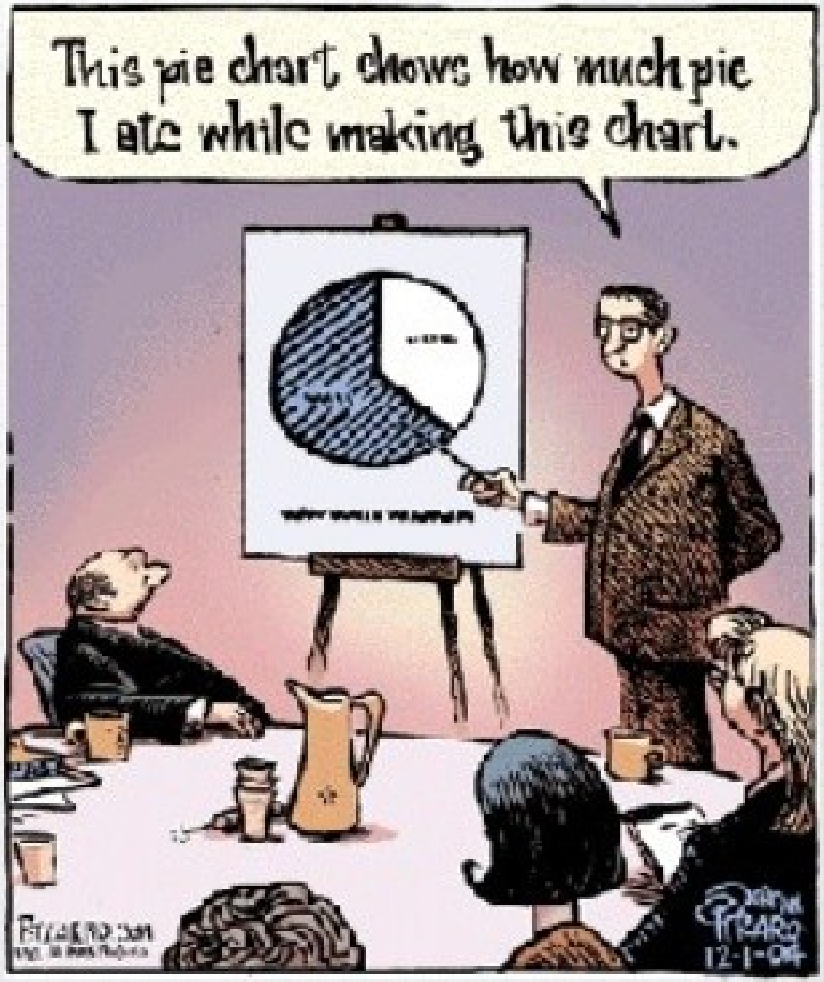
Data Analysis

Understanding your results by making a graph to analyze your data

Every graph tells a story



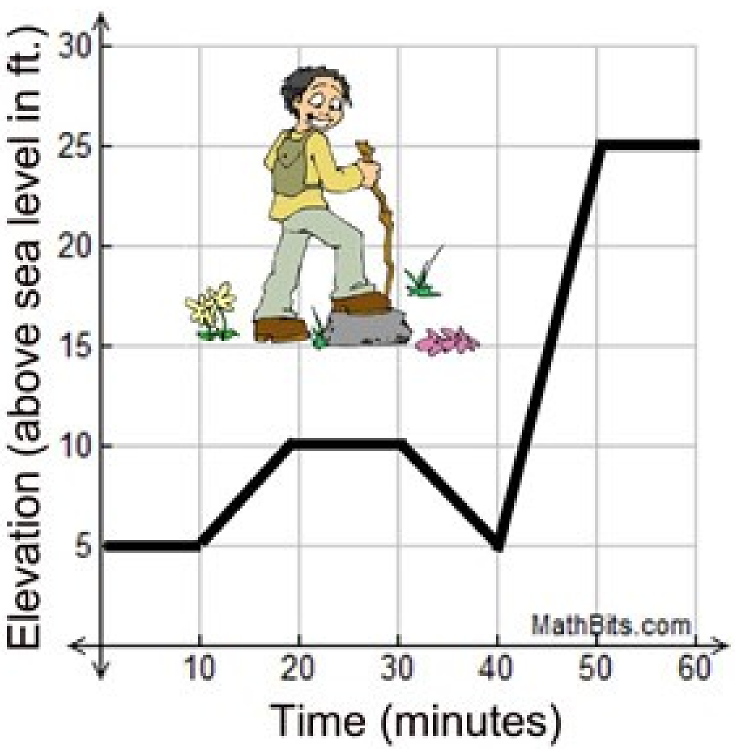
Goals for today

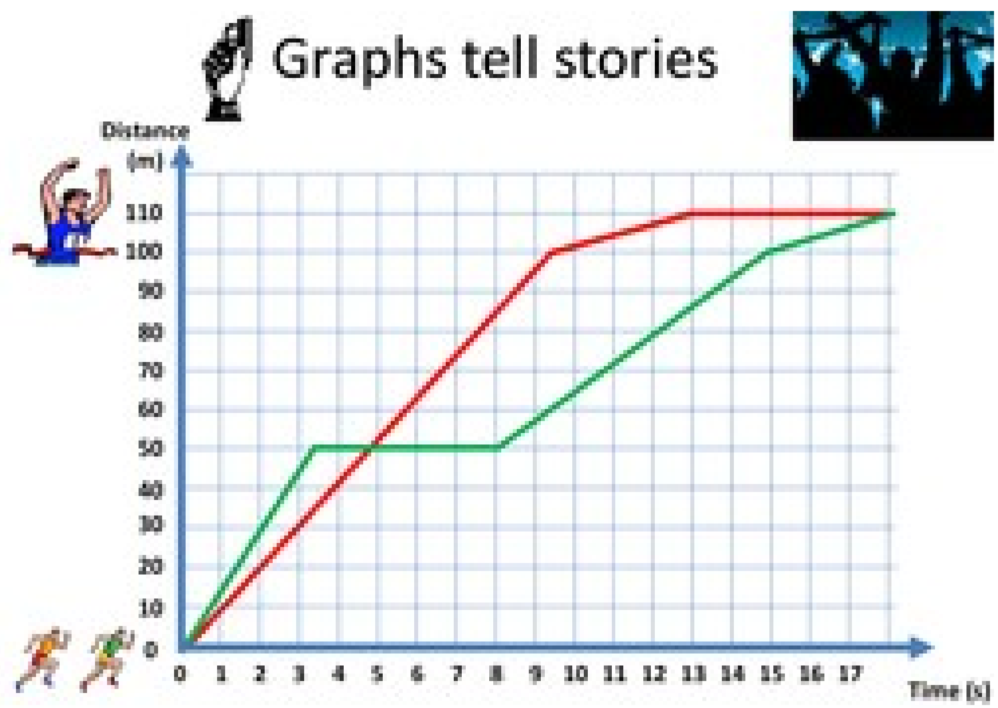
- Learn how to make a Graph from data you have collected
  - Data tables, labels, type
- Learn how to describe a graph
  - linear, non-linear, exponential
- Learn how to interpret a graph
  - What is the relationship? – proportional, directly proportional, inversely proportional, negative relationship, positive relationship
  - What physical quantity does the slope tell you?

Why Use Graphs

Organizing our data into a graph does more than make it aesthetically pleasing (pretty)

It tells a story





Data Tables

Make a data table and then use it to plot your data

X	Y
0	0
1	2
2	4
3	6

IV	DV
0	0
1	2
2	4
3	6

Time (seconds)	Distance (meters)
0	0
1	2
2	4
3	6

t (s)	d (m)
0	0
1	2
2	4
3	6

GENERAL ---SPECIFIC

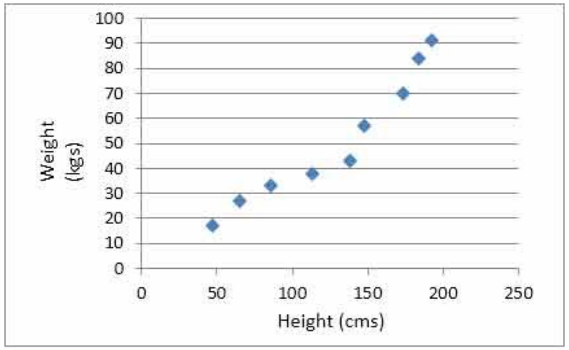
MATH---SCIENCE

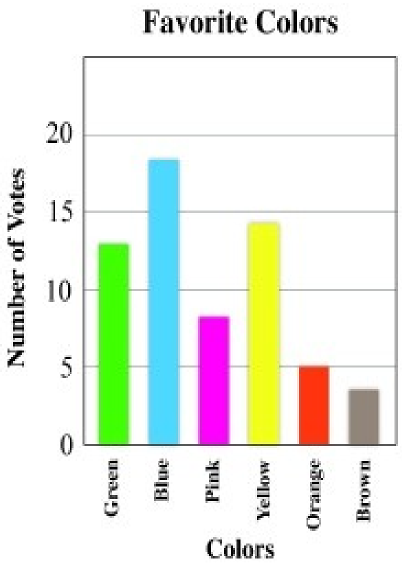
Graph Types

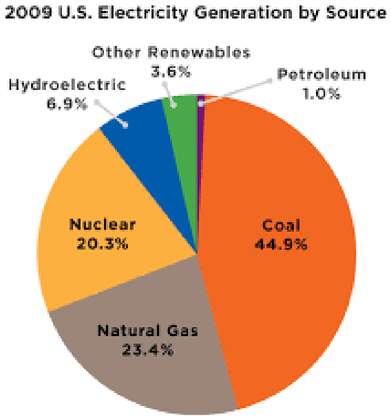
X-Y Scatter Plot

Bar/Column Graph

Pie Chart

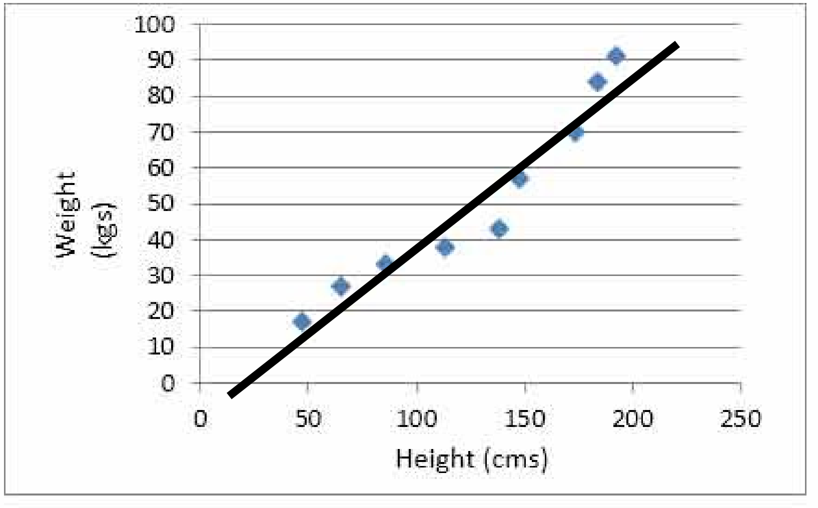






X-Y Scatter Plot

- used to determine **relationships** between the two different variables
- X-axis – Independent Variable
- Y- axis – Dependent variable
- most common type of graph you will use in science
- Do not connect the dots, instead use a trendline!

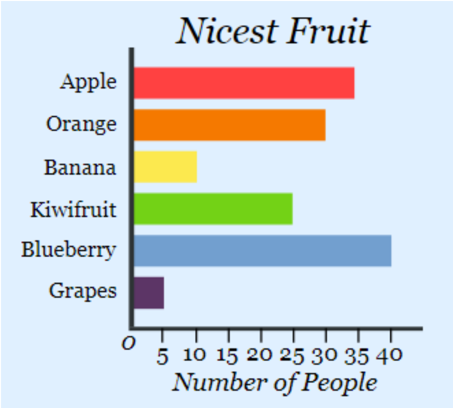


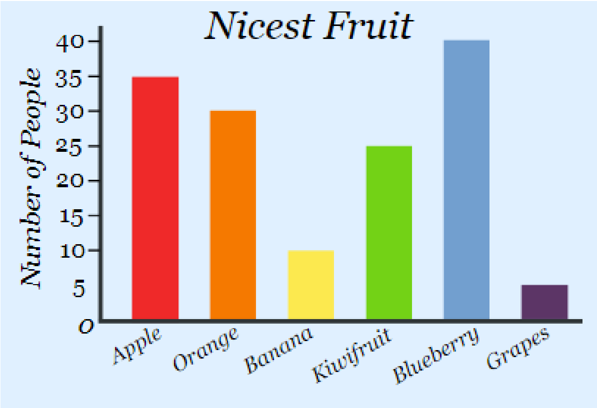
Bar/Column Graph

used to **compare** two or more different groups

they do not show changes over time

There is no relationship

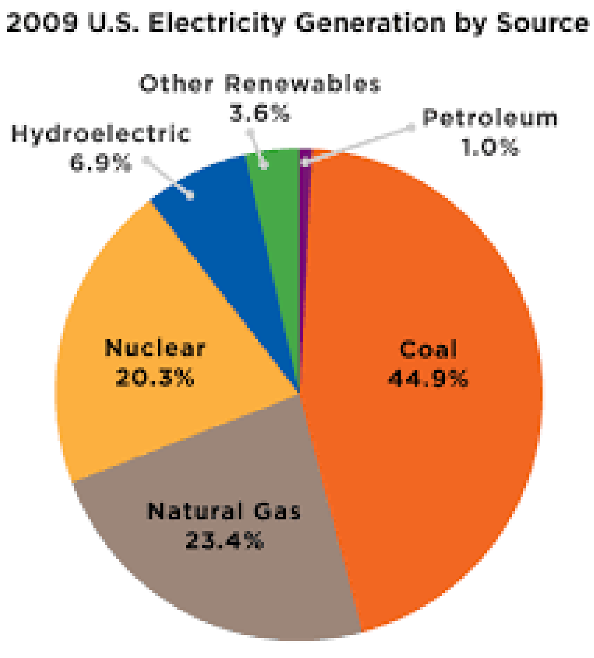




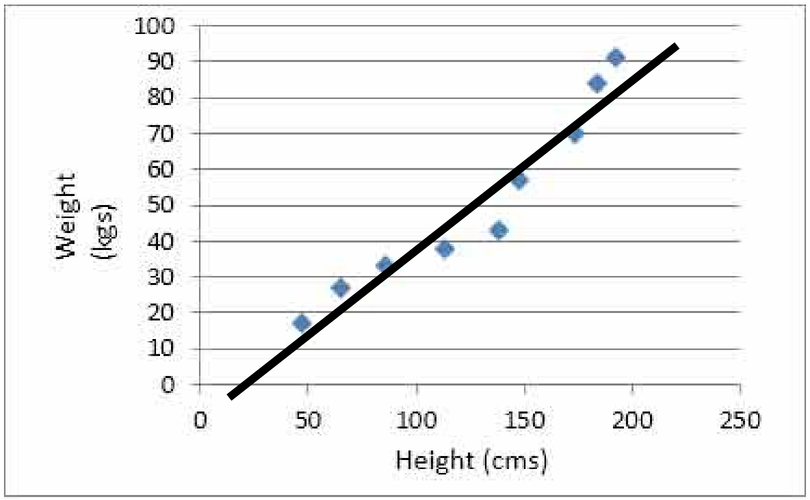
Both the **Bar** and the **Column** charts display data using rectangular **bars** where the length of the **bar** is proportional to the data value. Both are used to compare two or more values. However, their **difference** lies in their orientation. A **Bar chart** is oriented horizontally whereas the **Column chart** is oriented vertically.

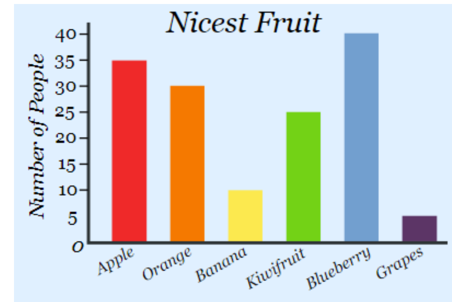
Pie Chart

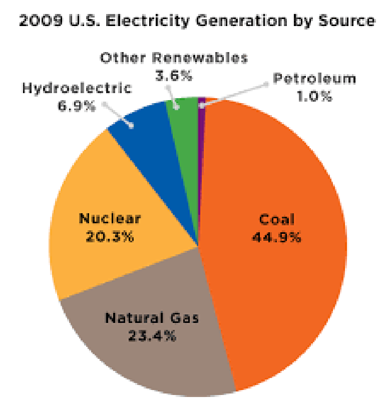
used when you are trying to compare parts of a whole
they do not show changes over time
There is no relationship



Where is the Relationship Shown?







RELATIONSHIP BETWEEN Y-VALUE AND X-VALUE

NO RELATIONSHIP

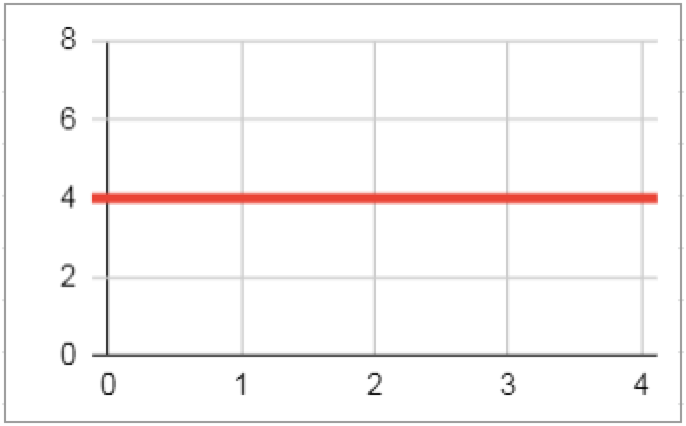
How to Interpret an X-Y Scatter Plot

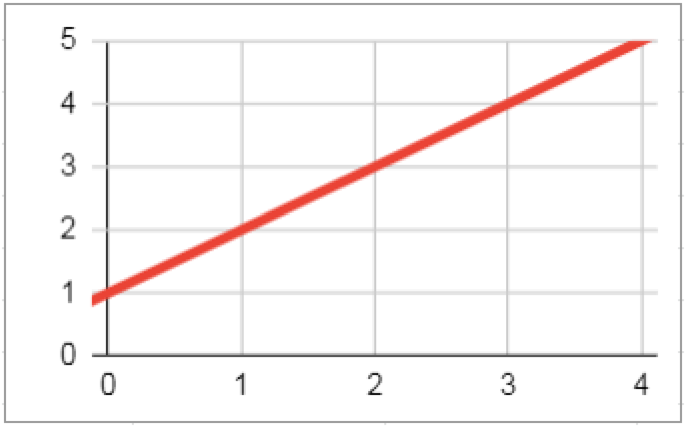
Is it:

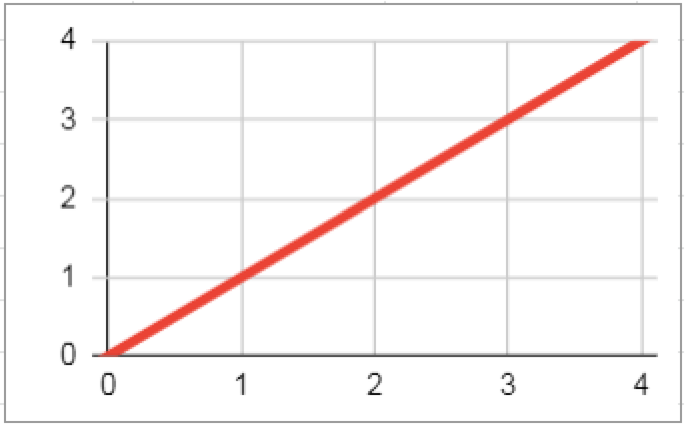
- Linear or non-linear
- Proportional, directly proportional, or inversely proportional
- Positive or negative relationship

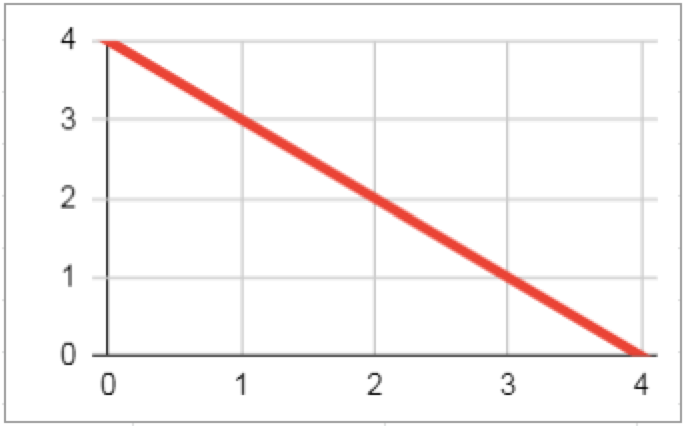
Linear & Constant Slope

These types of graphs will always have straight lines



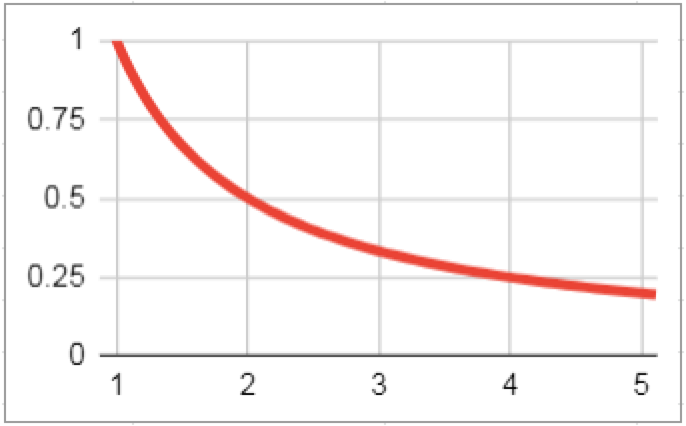


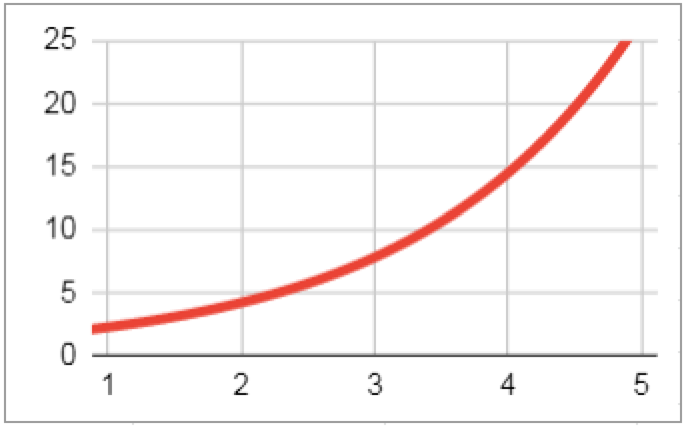




Non Linear & Changing Slope

These types of graphs will always have curved lines

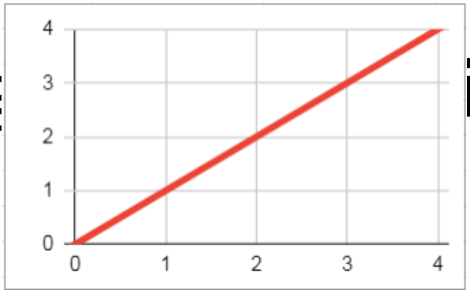


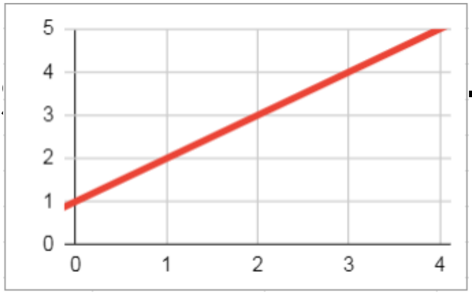


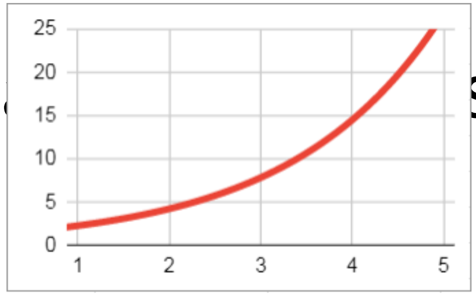
Positive or Negative Relationship

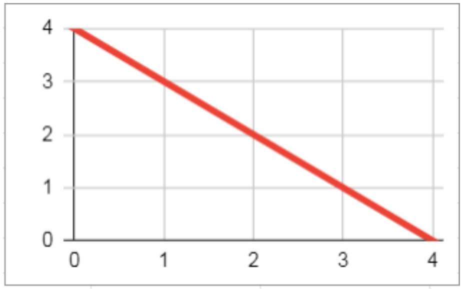
Positive - as x-axis increases, y-axis increases too

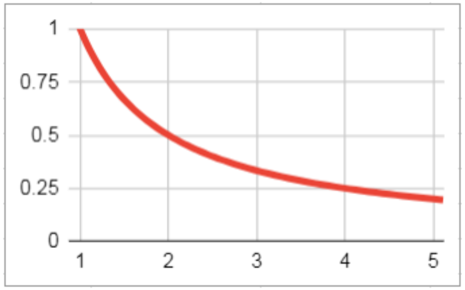
Neg









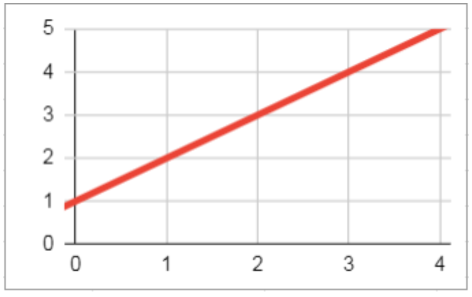


Proportionality

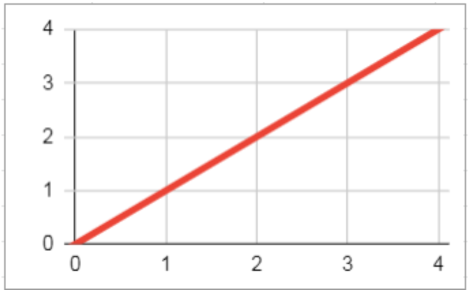
Proportional – as IV increases, DV increases

Directly Proportional – as IV increases, DV increases and goes through 0,0

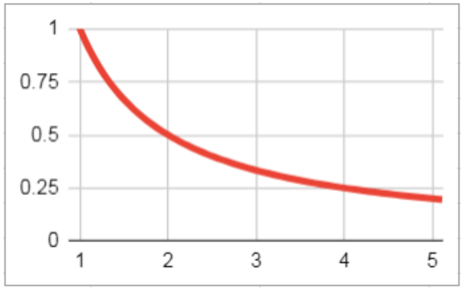
Inversely Proportional – as IV increases, DV decreases



PROPORTIONAL



DIRECTLY PROPORTIONAL



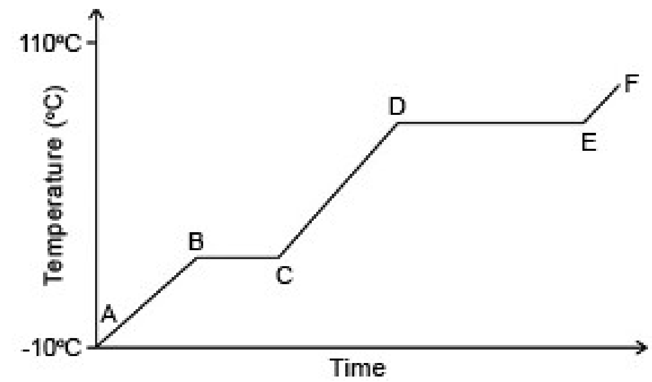
INVERSELY PROPORTIONAL

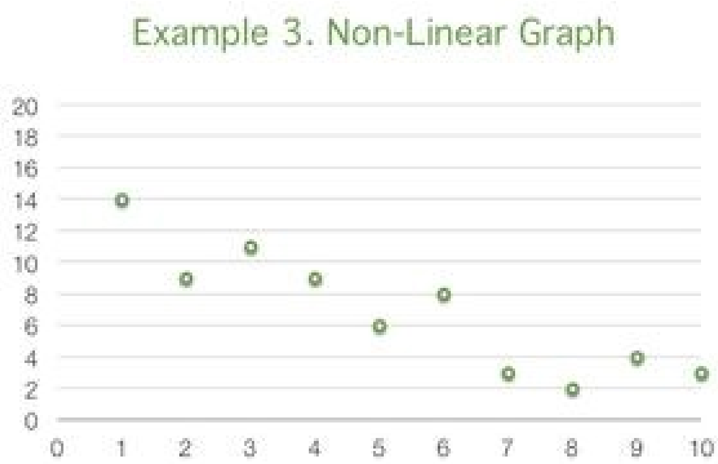
Nonlinear

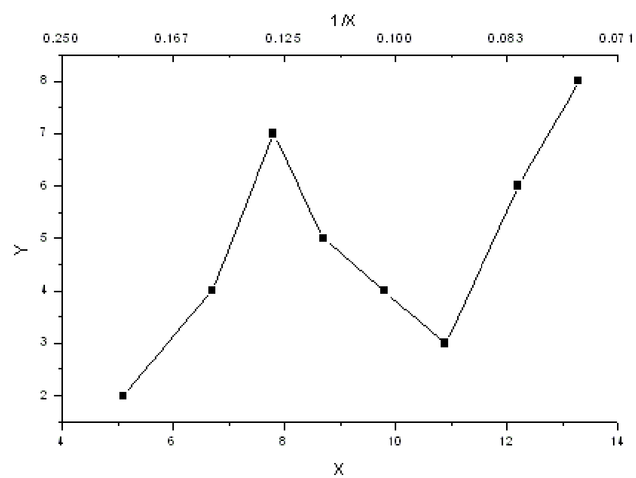
Not a straight line path

Look at the WHOLE picture for trends or patterns

may or may not be present

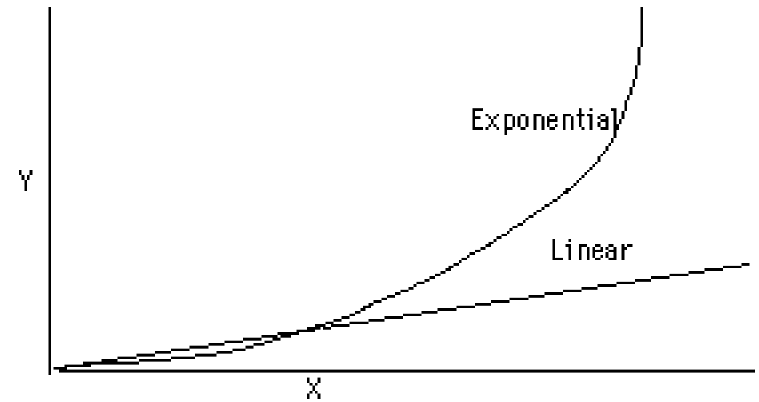


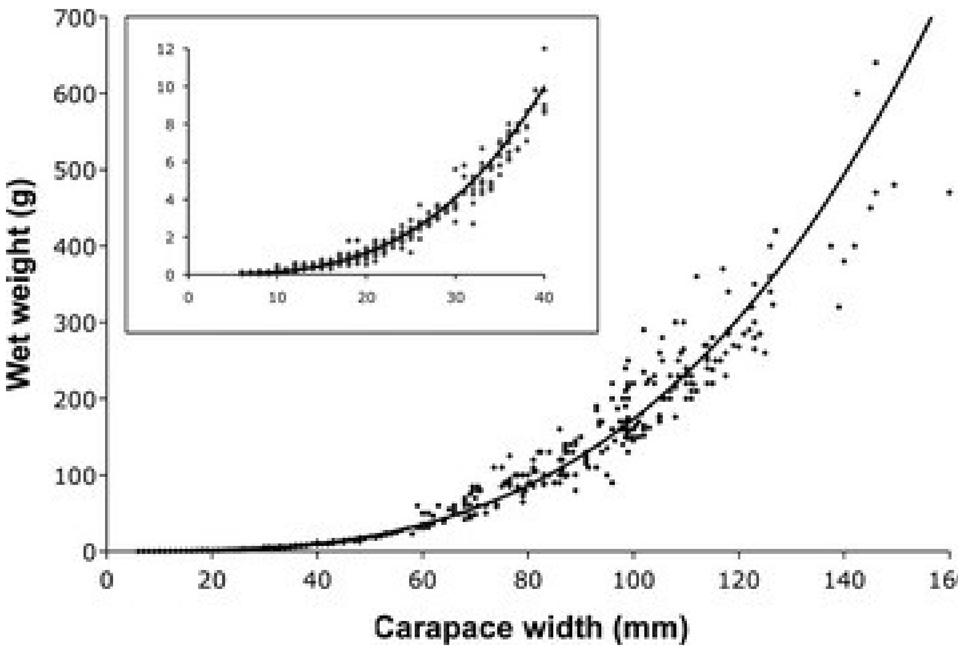


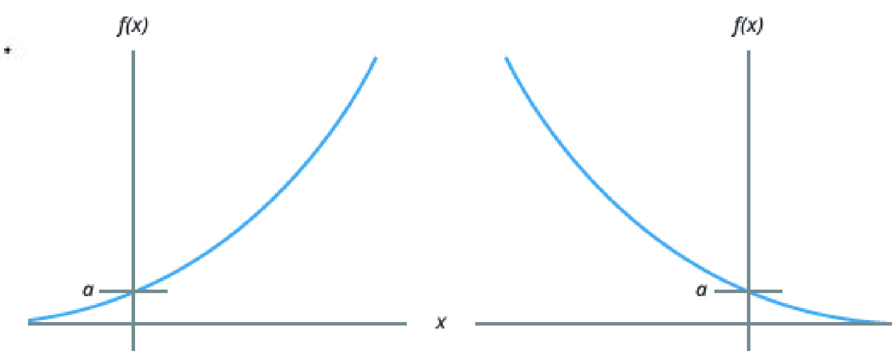


Exponential

- Graph increases or decreases by huge factors (shaped as a curve)







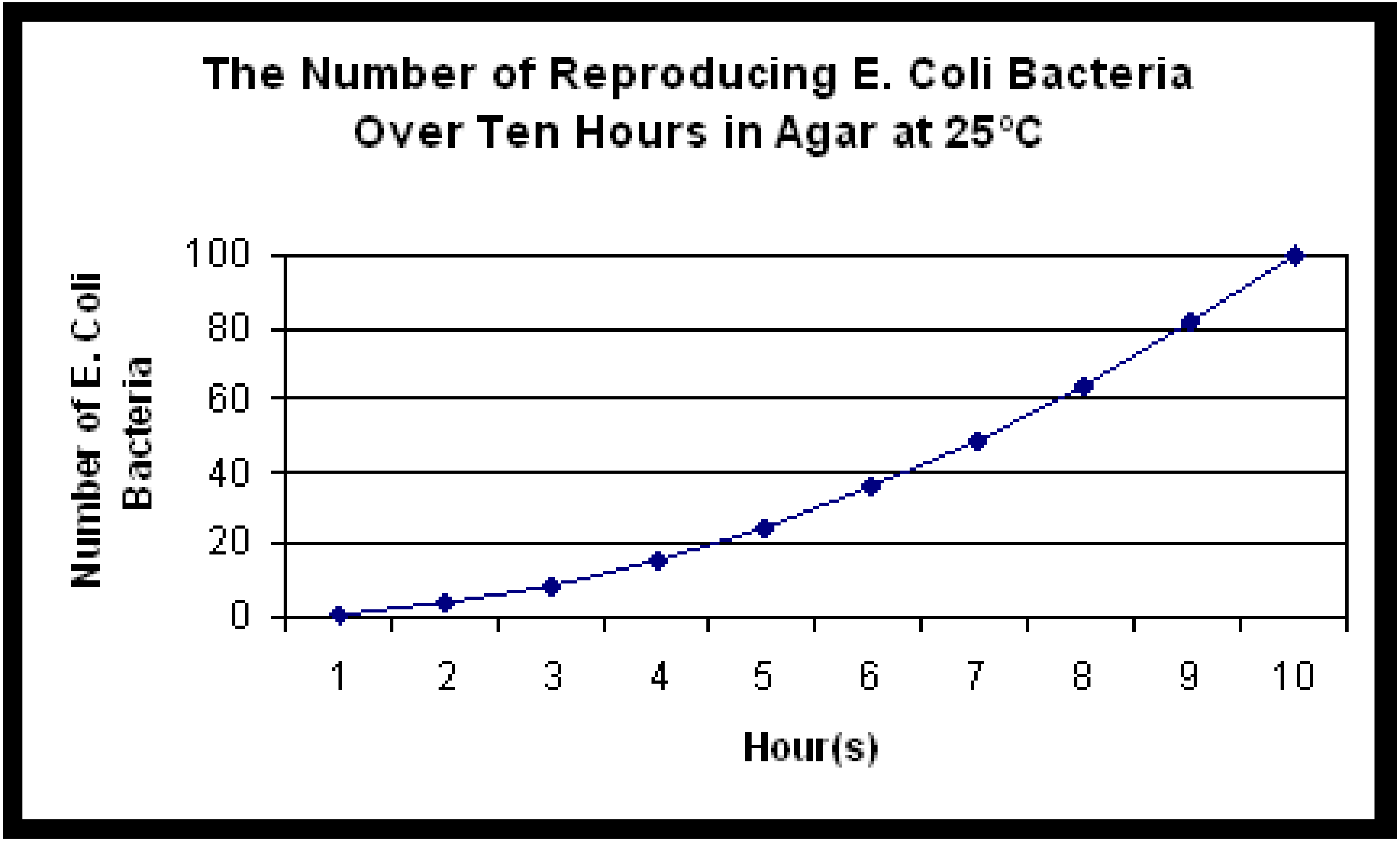
(a) Exponential growth
$f(x) = ab^x$ with $b > 1$

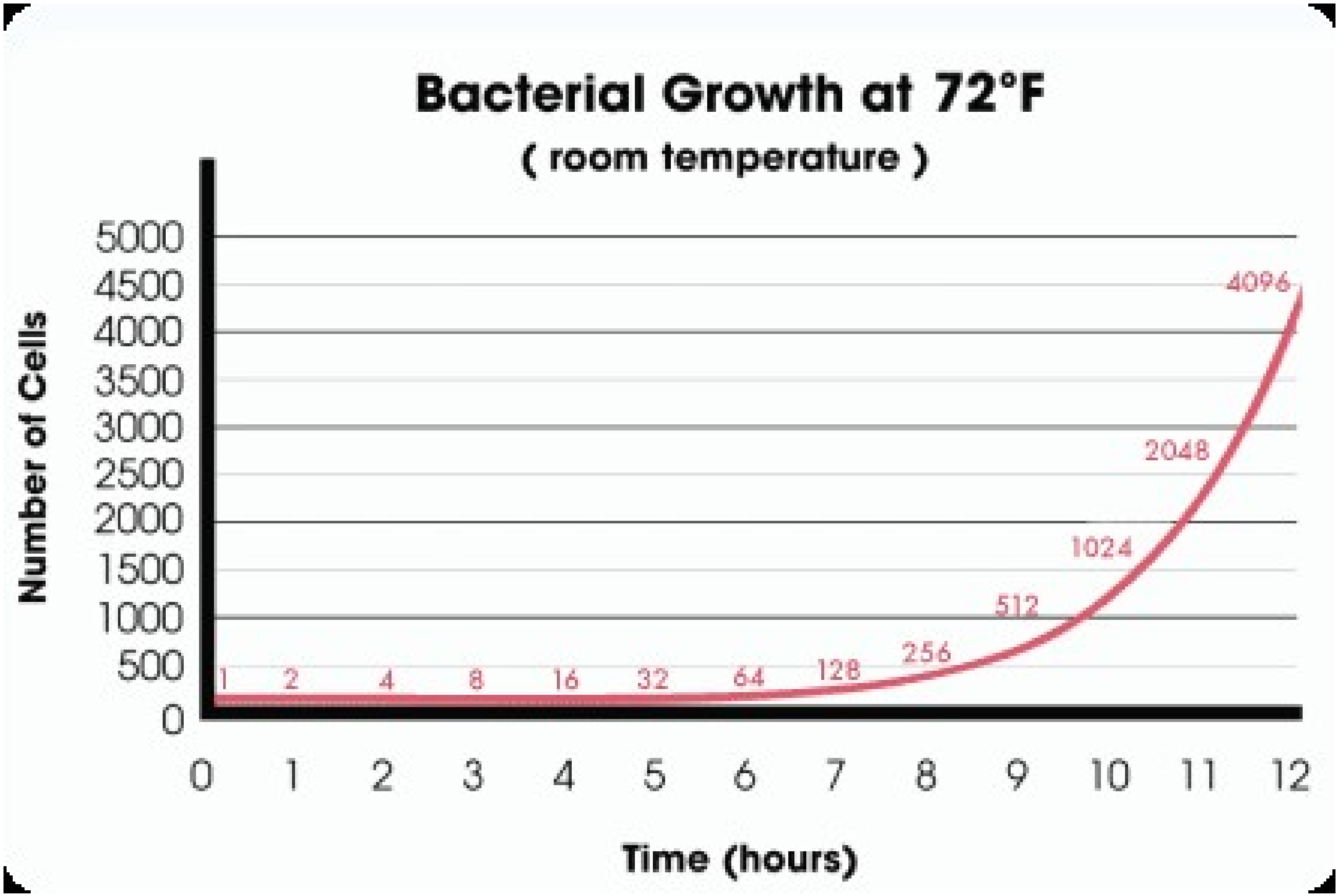
(b) Exponential decay
$f(x) = ab^x$ with $0 < b < 1$

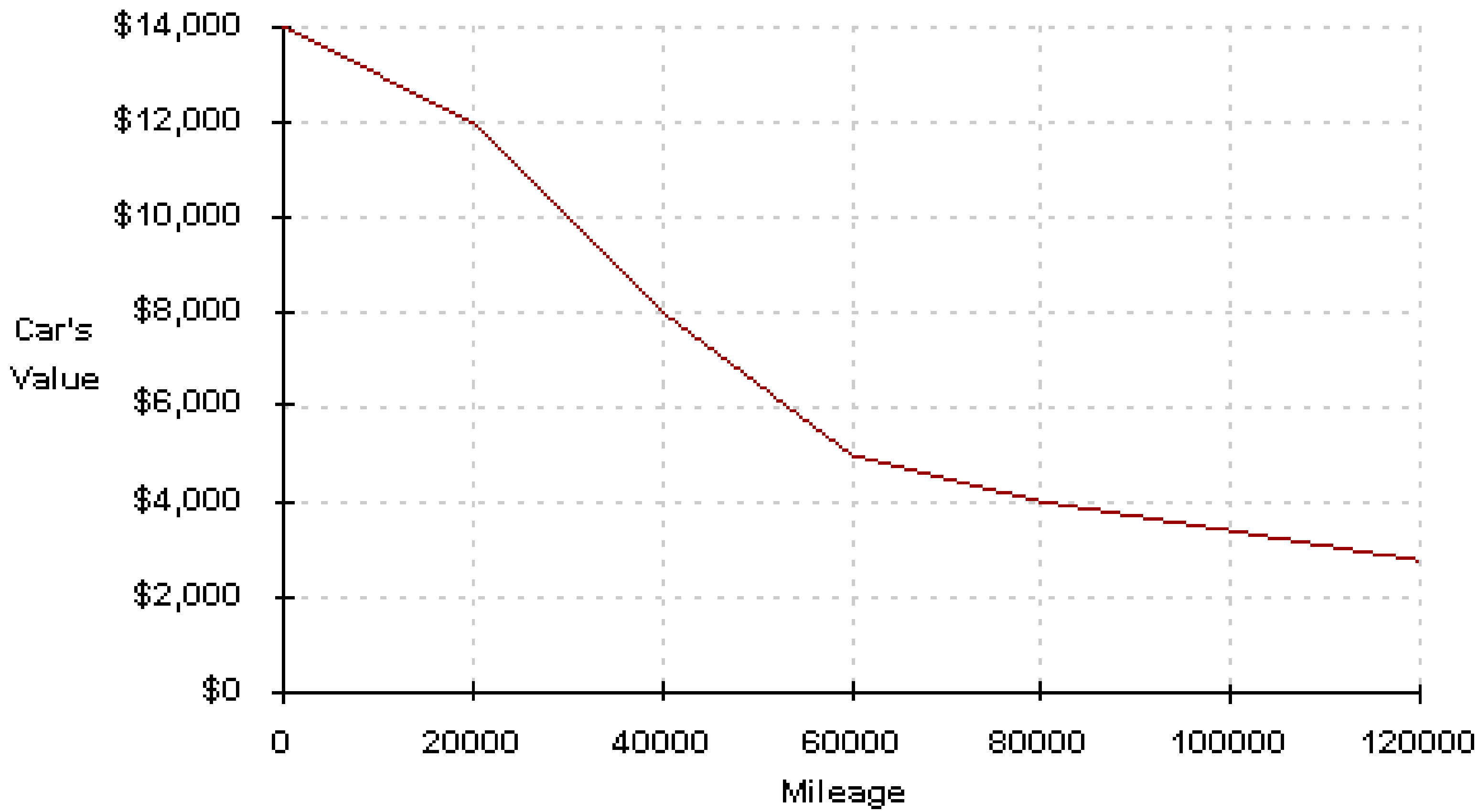
Examples

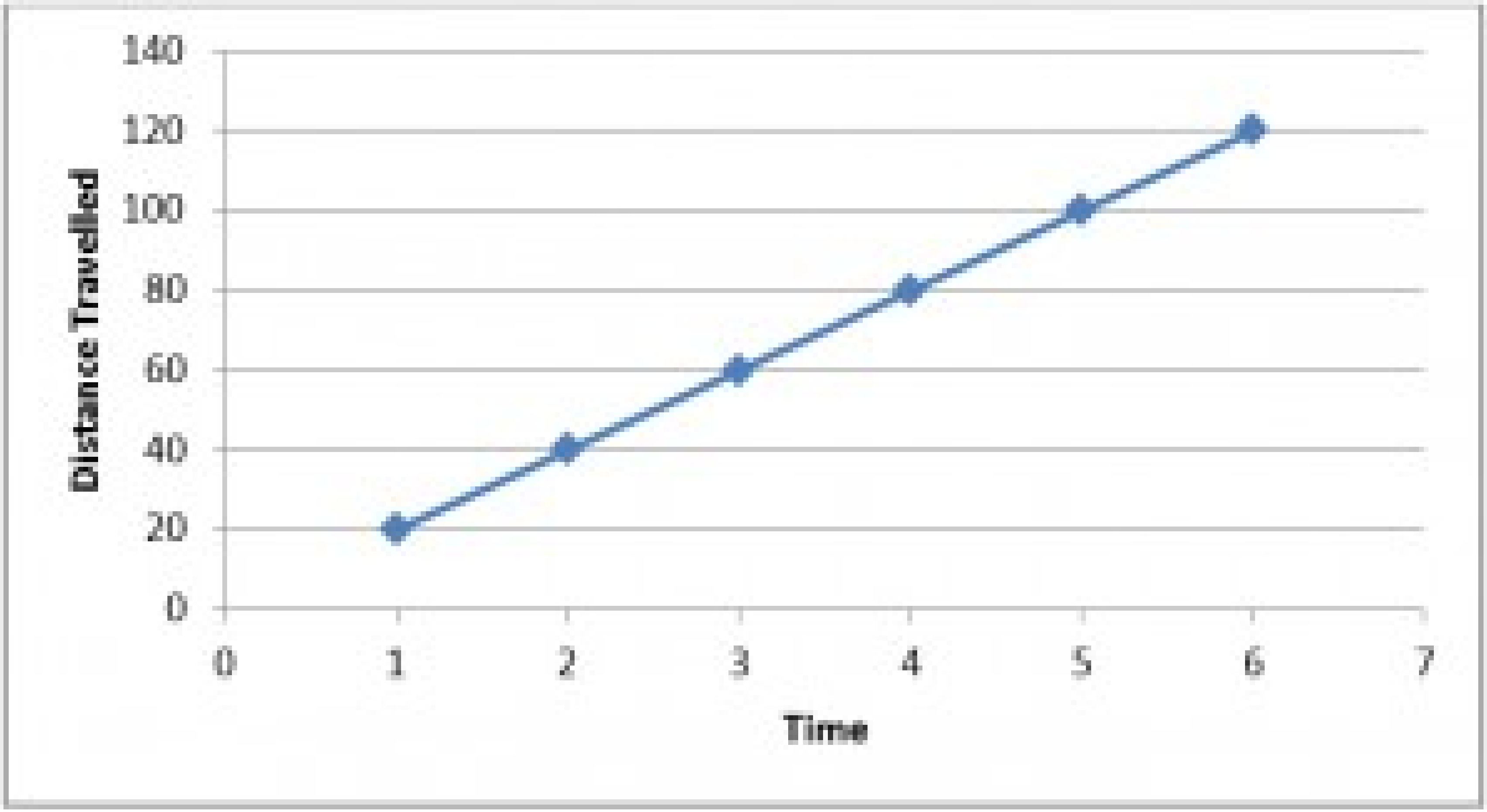
Discuss the following graphs

- What shape is the graph? (linear, nonlinear, exponential)
- What is the relationship between variables? (positive or negative)
- Is it proportional, directly proportional, inversely proportional?
- What story does it tell? (as X increases, Y ________)


The Number of Reproducing E. Coli Bacteria
Over Ten Hours in Agar at 25°C
Number of E. Coli Bacteria
100
80
60
40
20
0
1
2
3
4
5
6
7
8
9
10
Hour(s)


Bacterial Growth at 72°F
(room temperature)
Number of Cells
5000
4500
4000
3500
3000
2500
2000
1500
1000
500
0
1
2
4
8
16
32
64
128
256
512
1024
2048
4096
0
1
2
3
4
5
6
7
8
9
10
11
12
Time (hours)


$14,000
$12,000
$10,000
$8,000
$6,000
$4,000
$2,000
$0
Car's Value
0
20000
40000
60000
80000
100000
120000
Mileage



140
120
100
80
60
40
20
0
Distance Travelled
0
1
2
3
4
5
6
7
Time

Making a Graph

Make a data table
Choose a graph type
Plot your data
Do not connect the dots
Add a trendline if appropriate
    If linear or exponential
Give your graph a title
Label your axes with variable and unit
Scale your axes

Make a data table

Put your data in two x-y columns and label it

x	*y*
−2	−8
−1	−6.5
0	−5
1	−3.5
2	−2

Depth (km)	Basalt melting temperature (°C)
0	1100
25	1160
50	1250
100	1400
150	1600

MATH CLASS

SCIENCE CLASS

ONLY WRITE UNITS IN TOP BOXES WITH LABELS

Label Your Scatterplots in this Manner

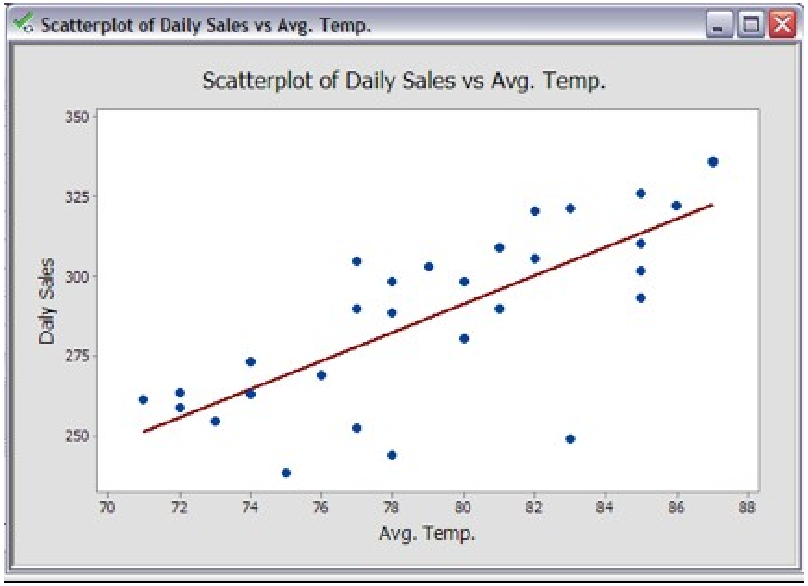
- Give your graph a descriptive title
  - “The relationship between DV (y-axis quantity) and IDV (x-axis quantity)”
- Label your axes with 3 things
  - What it is measuring
    - i.e. temperature, time, mass
  - Units of measurement
    - i.e. Celsius, seconds, grams
  - Show your scale
    - numbers

Add a Trendline

DO NOT CONNECT THE DOTS

Add a trendline, linear, exponential or power series as appropriate

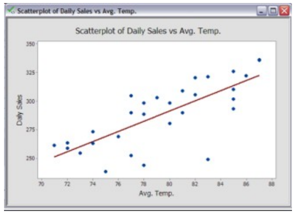
A trendline is the average of your data



Label Your Bar and Pie Charts in this Manner

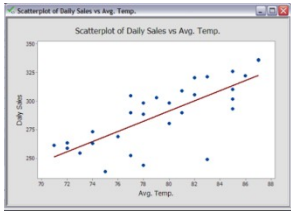
- Give your chart a descriptive title
  - DO NOT call it a relationship
  - Explain what the chart is telling you
  - “Comparison” might be a good title word
- For Bar/Column Charts
  - Label your axes appropriately
  - if necessary provide a key to identify what bars are telling you

Let’s Try it In Google Sheets!



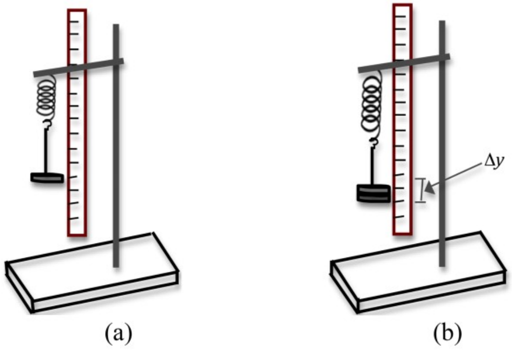
Everybody open a blank Google Sheets file and follow along with your instructor.

Let’s Plot a Relationship!

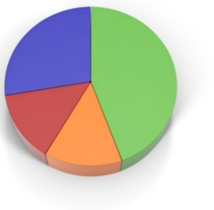


The Force (F) a spring exerts is directly proportional to the distance it is stretched and can be written in the form of a linear [y = f(x)] equation as F = kx, where k is the spring constant. Let’s make a scatterplot of F = kx for x = 0, .025, .045, .080 & .095 m where k equals 100.

F= 100x



Let’s Make a Pie Chart!



What percentage of students in this class have birthdays during the following quarters?

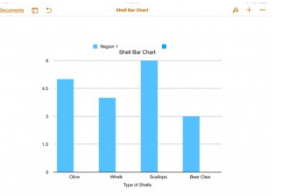
Q1 Jan-Mar- 25%

Q2 Apr-Jun-10%

Q3 Jul-Sep- 5%

Q4 Oct-Dec-60%

Let’s Make a Bar/Column Graph



Compare the different cars at the car rental agency when the following cars are available for rent:

10 Jeeps, 25 SUVs, 15 Economy Cars, 25 mid-size cars, 5 luxury cars, and 15 convertibles.